

# Kanwarpal Brar

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## Education

### University of Waterloo

Waterloo, Ontario, Canada

Bachelor of Computer Science (BCS) | CGPA: 3.84/4.0

2020 - 2025

**Courses:** Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Distributed Systems, Concurrency, Databases

## Work Experience

### Meta

#### Production Engineer

Sept 2025 - Present

- Accelerated **Meta Blob Storage** turnup **95%** via typed schema automation, unblocking rapid capacity delivery for **AI training** fleets
- Saved **100+** engineering hours by shipping **key-value** state layer atop **Infrastructure-as-Code** definitions, onboarding legacy systems
- Restored cluster capacity in **24 hours** by diagnosing cross-stack **garbage collection SEV**, unblocking environments for **6** downstream services
- Safeguarded **95% SLO** across **7** MBS services as primary **oncall**, protecting **AI training** pipelines for **Meta Superintelligence Labs**

### Carta

#### Payments Software Engineering Co-op

Sept 2024 - Dec 2024

- Averted **\$5000+** in regulatory penalties within first **2 weeks** by implementing financial compliance controls with **Django + React**
- Managed **\$5M+** monthly leading international banking integration expansion, leveraging **Python, gRPC**, and **Domain Driven Design**
- Architected fault-tolerant **microservices** on **AWS** with **Docker** and **Kubernetes**, achieving **99.99%** uptime for robust fintech operations
- Slashed network overhead **50%** by implementing centralized money movement controls with **RBAC** in a **Java Microservice**
- Accelerated feature development **40%** through flexible Django permission system using YAML-templated definitions

### University of Waterloo

#### Distributed Systems Research Assistant

May 2024 - Aug 2024

- Benchmarked **serverless** frameworks with **WRK** and custom **Bash** scripts, informing design decisions for high-performance system architecture
- Delivered **200+** requests/second per node by deploying & optimizing **Kubernetes** clusters with **KNative** Serving and **Istio** ingress controls
- Cut cold-start latency **20%** through targeted autoscaling and **TCP/IP** network optimization, increasing platform responsiveness
- Designed architectural changes improving resource utilization **15%**, validated with monitoring scripts in a comprehensive technical report

### Carta

#### Backend Software Engineering Co-op

Jan 2024 - Apr 2024

- Accelerated OCX report generation speed **10%** using an O(n) complexity cell management system with **Apache POI + Java**, streamlining results
- Improved semantic search relevance by **6%** via refined prompt engineering for report queries & real-world use-cases
- Eliminated **3000+** tickets by streamlining Ownership Report permissions in **Django**, collaborating cross-functionally with product leaders
- Slashed user wait times by **10%** through optimizing ownership report queries in **Spring**, enhancing system responsiveness

### Arctic Wolf Networks

#### Software Developer Co-op

May 2023 - Aug 2023

- Slashed response times by **25%** by implementing a concurrent **Go** monitoring system for **Prometheus** metrics on an **Apache Kafka** wrapper
- Attained **100%** accuracy in identifying test gaps with a reflection-based **Golang** unit test verifier for an **Apache Kafka** wrapper
- Expedited development by creating a forward/backward compatible Kafka Serialization system using **SchemaVer** and **Avro**

### Genesys Cloud Services

#### Full Stack Software Developer Co-op

Sept 2022 - Dec 2022

- Slashed hosting costs by **5%** by migrating a monolithic scheduling API to an end-to-end serverless architecture with **Python + Flask + Lambda**
- Compressed codebase size by **30%** and enhanced maintainability through a documented **Python + Flask + OpenAPI** service for scheduling

## Projects

### Event-Driven Data Aggregation App 🔄

2025

- Architected robust backend with **Go + Gin**, implementing **CQRS** and **Event Sourcing** for real-time data updates
- Achieved **sub-200ms** query latency with **Goroutines**, optimized **Event Sourcing** architecture, and efficient **WebSocket** handlers
- Crafted responsive cross-platform experience with **React Native + Google Maps SDK**, enabling dynamic location-based features

### Concurrent Lock-Free Hash Table 🔄

2025

- Built concurrent hash table in **C++** reaching **100M+** ops/sec on **64+** cores with **<1µs** latency using **lock-free** techniques and open addressing
- Achieved **10x** performance over mutex-based tables with **linear** scaling across cores using **CAS** operations and cache-line padding

## Skills

|                        |                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>      | Python, C++, Java, Go, TypeScript, JavaScript, Kotlin, C, SQL                                                               |
| <b>Frameworks:</b>     | Spring Boot, Django, Flask, Gin, Node.js, Express.js, React, React Native, GraphQL, REST APIs, OAuth2.0                     |
| <b>Data Systems:</b>   | PostgreSQL, MongoDB, Redis, Elasticsearch, Hadoop, Spark, HBase, Kafka, ZooKeeper, gRPC, Protocol Buffers, Avro             |
| <b>Infrastructure:</b> | AWS, Azure, Kubernetes, Docker, Terraform, CI/CD, Linux, Git, Microservices, CQRS, Event Sourcing, TCP/IP, eBPF, Prometheus |
| <b>AI / GenAI:</b>     | LLMs, Generative AI, RAG, Prompt Engineering, LLM Evals, Agents, Claude Code, MCP, LangChain                                |